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Effects of fat content of high-protein milk replacer on plasma metabolite and hormone concentrations of Holstein calves in summer and winter

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ABSTRACT

The objective of this study was to evaluate the effects of fat content of high-protein milk replacer on plasma metabolite and hormone concentrations of dairy calves in summer and winter. Eighty-four Holstein heifer calves (BW at 7 d of age, 42.2 ± 2.54 kg; mean ± SD) were assigned to 1 of the 3 treatments: milk replacer containing 29% CP and 21% fat (LF; 4.7 Mcal of ME/kg), 26% fat (MF; 5.0 Mcal of ME/kg), and 32% fat (HF; 5.3 Mcal of ME/kg) on a DM basis (n = 14 each for summer and winter). Milk replacers were offered at 600 g/d (powder basis; 3.6 L/d) from 8 to 14 d, increased up to 800 g/d (4.8 L/d) from 15 to 21 d, 1,200 g/d (7.2 L/d) from 22 to 42 d, decreased down to 800 g/d (4.8 L/d) from 43 to 49 d, and 600 g/d (3.6 L/d) from 50 to 56 d, then weaned at 56 d of age. Data and samples were collected until 91 d of age. All the calves were fed a calf starter (23% CP) and chopped hay (11% CP) ad libitum from 7 d of age. Increasing the fat content of milk replacer decreased plasma urea nitrogen concentrations during the peak and weaning periods. Additionally, we observed an interaction of treatment and season for plasma urea nitrogen concentration during the step-up period; it was decreased with the fat content of milk replacer increased in winter but was not affected by treatment in summer. Calves had higher plasma total cholesterol and fatty acids concentrations with increased fat content of milk replacer during the preweaning period. Plasma insulin concentration was differently affected between summer and winter; the concentration was higher in the calves fed the 26% fat treatment than the calves fed the 21% fat treatment in the winter, but in summer, it was not affected by treatment and showed high concentration overall. In conclusion, increasing the fat content of milk replacer decreased plasma urea nitrogen concentration during the peak and weaning periods regardless of season, but only in winter during the step-up period. In addition, greater fat content of milk replacer increased plasma total cholesterol and fatty acid concentrations during the preweaning period regardless of season, but did not exert consistent effects on plasma insulin concentration.

Key words: calf, milk replacer, fat, thermal stress, metabolism

INTRODUCTION

Recent research on calf nutrition recommends that milk replacer for calves contain more fat to approximate the macronutrient profile of bovine whole milk (Amado et al., 2019; Wilms et al., 2022; Ghaffari et al., 2024). Fat content of milk replacer affects nutrient metabolism. Regarding fat metabolism, previous studies reported that high-fat milk replacers containing 23% to 31% fat had higher blood total cholesterol and fatty acids concentrations compared with those containing 17% to 21% fat (Berends et al., 2020; Echeverry-Munera et al., 2021; Wilms et al., 2022). These seemed to be a direct result of the higher fat content of milk replacer. Regarding nitrogen metabolism, increasing the fat content of milk replacer was reported to decrease (Berends et al., 2020) or increase (Echeverry-Munera et al., 2022) BUN concentrations; results differ between studies. When increasing the fat content of milk replacer, lactose content is reduced in contrast (Amado et al., 2019; Echeverry-Munera et al., 2021; Wilms et al., 2022). Low-fat (17%–21% DM) and high-lactose (42%–44% DM) milk replacer has been linked to impaired glucose homeostasis, including hyperglycemia, glucosuria, and hyperinsulinemia (Hugi et al., 1998; Echeverry-Munera et al., 2021). Welboren et al. (2021) suggested that lower lactose inclusion in milk replacer, in other words greater fat inclusion could be beneficial for glucose homeostasis because smaller fluctuations in postprandial glucose and insulin concentrations have been observed.

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

Table 1. Nutrient composition of milk replacers, calf starter, and Klein grass hay

Item	Milk replacer[1] LF	MF	HF	Calf starter	Klein grass hay
Ingredients of milk replacers					
Dried skim milk	50	50	50	—	—
Whey protein (mid-protein)	22	22	11	—	—
Whey protein (high-protein)	0	1	6	—	—
Dried whey	7	0	0	—	—
Soy protein concentrate	2	2	2	—	—
Vegetable fat	17	23	28	—	—
Others	2	2	2	—	—
DM, %	95.7	95.9	96.1	88.8	92.3
Nutrient composition, % of DM					
CP	29.1	28.8	28.9	22.6	11.1
Fat	20.5	26.3	31.7	3.5	2.0
Fatty acids	17.8	23.5	28.8	—	—
Ash	7.2	6.7	6.3	6.4	8.3
NDF	0.4	0.2	0.3	16.9	63.3
NFC	42.8	38.0	32.8	50.6	15.3
Starch	0.1	0.1	0.1	34.1	0.4
Lactose	39.9	35.3	30.2	—	—
Water soluble carbohydrate	—	—	—	10.1	6.2
ME[2], Mcal/kg	4.69	4.98	5.26	3.29	2.07
Osmolality,[3] mOsm/kg	478	424	376	—	—
Fatty acid composition, % of total fatty acid					
Butyric acid (C4:0)	0.3	0.2	0.1	—	—
Caprylic acid (C8:0)	3.5	3.6	3.6	—	—
Capric acid (C10:0)	3.3	3.3	3.3	—	—
Lauric acid (C12:0)	12.2	12.5	12.6	—	—
Myristic acid (C14:0)	4.9	4.7	4.6	—	—
Palmitic acid (C16:0)	29.2	29.2	29.2	—	—
Stearic acid (C18:0)	4.2	4.1	3.9	—	—
Oleic acid (C18:1 n-9)	32.6	32.8	33.1	—	—
Linoleic acid (C18:2 n-6)	8.3	8.3	8.3	—	—
Alpha-linolenic acid (C18:3 n-3)	0.2	0.2	0.2	—	—
Arachidic acid (C20:0)	0.3	0.3	0.3	—	—
Other	1.0	0.9	0.8	—	—

[1]HF = milk replacer containing 32% fat; LF = milk replacer containing 21% fat; MF = milk replacer containing 26% fat.

[2]ME was calculated according to NASEM (2021).

[3]Osmolality was measured. mOsm = milliosmoles.

In addition, thermal stress affects blood metabolism and hormone of calves. Heat stress has been shown to enhance insulin secretion and sensitivity (Wang et al., 2020). Calves from noncooled heat-stressed dams had higher insulin concentrations compared with those from cooled dams the first 4 d after birth (Tao and Dahl, 2013). O'Brien et al. (2010) similarly observed that heat-stressed calves had higher insulin concentration. High insulin concentration from heat stress inhibited fat mobilization from the adipose tissue, thereby lowering plasma fatty acid concentrations in lactating cows (Rhoads et al., 2009). Consequently, with less body fat–derived energy available, protein catabolism in mammary glands and muscles are promoted to obtain more energy substrates (Baumgard and Rhoads, 2013).

However, previous studies did not consider the interaction between fat content of milk replacer and environmental temperature. From previous studies, we hypothesized that feeding high-fat milk replacer for calves results in different metabolic profiles in winter and summer. The objective of this study was to evaluate the effects of fat content of high-protein milk replacers on plasma metabolite and hormone concentrations of dairy calves in summer and winter.

MATERIALS AND METHODS

Animals and Housing

The experimental design and partial data of this experiment, which contains intake, growth performance, and health, were reported by Fukami et al. (2025). All experimental procedures were approved by the Animal Care and Use Committee of Hiroshima University (#20-2-2). The experiments were conducted from June to November (summer), and from November to March (winter). A total of 84 Holstein heifer calves (n = 42 each for summer and winter) were transported from commercial dairies to the Dairy Technology Research Institute (Yabuki, Fukushima, Japan) at 3 to 5 d of age. The calves were born between June 8 and August 21, 2023 (summer), and October 26 and December 19, 2022 (winter). When calves arrived at the research farm, they were checked for respiratory disease and diarrhea, and calves with clinical symptoms were excluded from the study. Upon arrival, calves received 360 mg of tilmicosin (1.2 mL of Micotil 300 injection, Zenoaq, Fukushima, Japan), 11 mg of sodium selenite, and 100 mg of d-α-tocopherol acetate (2 mL of ESE, Kyoritsu Seiyaku, Tokyo, Japan), combination vaccine (2 mL of TSV-3, Zoetis Japan, Tokyo, Japan) and 25 mg of Ivermectin (5 mL of Ivelmec PO, Fujita Pharm, Tokyo, Japan) via subcutaneous injection, intramuscular injection, nasal administration, and percutaneous absorption, respectively. In addition, calves received 300,000 IU of vitamin A, 30,000 IU of vitamin D_3, and 120 mg of d-α-tocopherol acetate (3 mL of Vitalap 101, Zenoaq, Fukushima, Japan) via oral administration. The calves were fed 2 L of electrolyte solution on the day of arrival, and a milk replacer (29% CP and 26% fat on DM basis) up to 600 g/d (powder basis) thereafter until a.m. of d 7. At the beginning of the study (7 d of age), BW was 42.2 ± 2.54 kg, and serum IgG concentration was 19.6 ± 13.5 g/L (mean ± SD). The calves were raised outdoors in individual hutches (115 cm × 230 cm × 120 cm) bedded with sawdust on a rubber mat throughout the study.

Feeding

The calves were blocked by birth date, BW, and farm origin and randomly assigned to one of the 3 treatments: milk replacer containing 21% fat (**LF**; 4.7 Mcal of ME/kg), 26% fat (**MF**; 5.0 Mcal of ME/kg), and 32% fat (**HF**; 5.3 Mcal of ME/kg) on a DM basis (n = 14 each for summer and winter). The LF contained 21% fat, which was a concentration used in high levels of nutrient intake for high ADG in the thermoneutral zone (Murayama et al., 2023). As the maintenance energy requirement increases at temperature out of the thermoneutral zone (15 to 25°C before 21 d of age, 10 to 25°C after 22 d of age; NASEM, 2021), we increased the energy content of milk replacers for MF and HF groups to provide additional energy potentially required for maintenance. All experimental milk replacers contained 29% CP on a DM basis were prepared at a concentration of 166.7 g/L, and offered at 600 g/d (powder basis; 3.6 L/d) from p.m. of d 7 to a.m. of d 14, 800 g/d (4.8 L/d) from p.m. of d 14 to a.m. of d 21, 1,200 g/d (7.2 L/d) from p.m. of d 21 to a.m. of d 42, 800 g/d (4.8 L/d) from p.m. of d 42 to a.m. of d 49, and 600 g/d (3.6 L/d) from p.m. of d 49 to a.m. of d 56, then weaned. Data and samples were collected until 91 d of age. The experimental period was divided into 4 phases: step-up period (8–21 d of age), peak period (22–42 d of age), weaning period (43–56 d of age), and postweaning period (57–91 d of age). All the calves were fed one of the milk replacers using a bucket with a soft rubber nipple twice daily at 0630 and 1615 h. All the calves were fed a texturized calf starter (22.6% CP and 16.9% NDF on a DM basis) and chopped klein grass hay (*Panicum coloratum* L.; 11.1% CP and 63.3% NDF on a DM basis) ad libitum separately at 1000 h from 7 d of age. The texturized calf starter contained steam-flaked barley grain, steam-flaked corn grain, beet pulp pellet, alfalfa dehydrated pellet, molasses cane, and the other ingredients in pellet. Klein grass hay was chopped to a theoretical length of 19 mm by a cutting machine (CX-201; Yamamoto Co. Ltd., Tendo, Yamagata, Japan). Analyzed nutrient compositions of the milk replacers, calf starter, and hay are in Table 1. The amount of refused milk replacer was recorded at each feeding. Refused calf starter and hay were removed daily at 1000 h, and their amounts were recorded. All the calves had free access to water.

Ambient Temperature

During the experiment, ambient temperatures were measured and recorded every 30 min using a thermometer data logger system installed in an empty hutch, at approximately 0.60 m above the ground. The ambient average, maximum, and minimum temperature were 23.6°C ± 2.6°C, 31.1°C ± 2.5°C, 18.5°C ± 2.6°C in summer. Ambient average, maximum, and minimum temperature were 0.9°C ± 2.7°C, 8.4°C ± 3.6°C, −4.1°C ± 2.4°C in winter. The number of days outside the thermoneutral zone during summer and winter was 47 d and 82 d, respectively per 84-d period.

Data and Sample Collection

Blood samples were collected 3 h after morning milk feeding (0930 h) at 7 d of age and weekly thereafter, using vacutainers for the collection of plasma (Venoject II VP-H100K with heparin sodium; Terumo Corporation) from the jugular vein. Blood samples continued to be collected at the same time (0930 h) during the postweaning period (63–91 d of age). Immediately after sample collection, aprotinin (500 kallikrein inhibitor units/mL of blood; Sigma-Aldrich Inc.) was added to the blood samples, centrifuged at 1,800 × *g* at 4°C for 20 min, and the plasma was collected. Plasma samples were stored at −80°C until analysis.

Sample Analysis

Analyses of DM, ash, CP, fat, starch, NDF, water soluble carbohydrate and lactose in milk replacer, calf starter, and hay samples are described in Fukami et al. (2025), and their ME contents were calculated according to NASEM (2021). Osmolality of the milk replacers was analyzed by a commercial laboratory (Japan Food Research Laboratories, Tokyo, Japan) using the Osmolarity Determination method from *The Japanese Pharmacopoeia* (MHLW, 2021). The fatty acid profile of the milk replacers was analyzed by a commercial laboratory (Japan Food Research Laboratories, Tokyo, Japan) using GC according to AOAC International (2002; method no. 996.06).

Plasma blood samples were analyzed for alkaline phosphatase, triglycerides, total cholesterol, fatty acids, albumin, total protein, urea nitrogen, glucose, total ketone body, IGF-1, and insulin concentrations. Plasma alkaline phosphatase, triglycerides, total cholesterol, fatty acids, albumin, total protein, urea nitrogen, glucose and total ketone body concentrations were determined via enzymatic methods using a clinical chemistry analyzer (Bio Majesty, JACK-BM, Jeol Ltd.). Plasma IGF-1 and insulin concentrations were measured using the time-resolved fluoroimmunoassay technique previously described by Sugino et al. (2004). Plasma IGF-1 concentrations were measured using europium-labeled human IGF-1 and polystyrene microtiter strips (Nalgene Nunc Int.) coated with goat anti-rabbit γ-globulin (Laarman et al., 2012). The plasma concentration of insulin was measured using a solid-phase competition immunoassay with europium-labeled bovine insulin and polystyrene microtiter strips coated with anti-guinea pig γ-globulin (Inabu et al., 2017). Intra- and interassay CV were 2.1% and 3.1%, respectively, and the least detectable level was 0.14 ng/mL.

Statistical Analysis

Empty body weight (**EBW**) was calculated from BW as 0.94 × BW from 8 to 21 d of age, 0.93 × BW from 22 to 56, and 0.85 × BW from 57 to 91 d of age according to NASEM (2021). Effects of thermal stress on additional maintenance requirements were calculated as 2.01 kcal/kg $EBW^{0.75}$ for each degree increase or decrease in environmental temperature (°C) above the upper or below the lower critical temperatures according to NASEM (2021).

Experimental design was a 3 × 2 factorial arrangement of treatments with 3 milk replacers (LF, MF, and HF) and 2 seasons (summer and winter). Plasma metabolites and hormones were analyzed separately for each nutritional phase (step-up, peak, weaning, and postweaning) using JMP Pro 19 (SAS Institute Inc., Cary, NC) with the following model:

$$Y_{ijkl} = \mu + M_i + S_j + D_k + MS_{ij} + MD_{ik} + SD_{jk} + MSD_{ijk} + C_l + e_{ijkl},$$

where Y_{ijkl} is the dependent variable, μ is the overall mean, M_i is the fixed effect of milk replacer treatment (i = 1, 2, and 3), S_j is the fixed effect of season (j = 1 and 2), D_k is the fixed effect of day as a repeated measure (k = 7, 14, …, 91), MS_{ij} is the interaction between milk replacer treatment and season, MD_{ik} is the interaction between milk replacer treatment and day, SD_{ik} is the interaction between season and day, MSD_{ijk} is the interaction among milk replacer treatment, season and day, C_l is the random effect of calves (l = 1, 2, …, 80), and e_{ijkl} is the residual. When significant or tendencies of treatment × season or treatment × season × day interactions were observed, treatment means were compared using Tukey's test separately for each season. Covariant structure of the model was compound symmetry.

Correlation coefficients were determined between energy requirement for thermal stress and plasma metabolic concentrations using Spearman's correlation method using JMP Pro 19 (SAS Institute Inc., Cary, NC). Significance was declared at $P \leq 0.05$, and tendency was declared at $0.05 < P \leq 0.10$.

RESULTS

Four calves experiencing severe diarrhea were excluded from the study. Therefore, LF and MF groups had 2 calves less than HF group; LF group has 26 calves (n = 12 for summer, n = 14 for winter), MF group has 26 calves (n = 12 for summer, n = 14 for winter), and HF group has 28 calves (n = 14 for summer, n = 14 for winter).

Feed and Nutrient Intake

The calves fed the HF treatment had lower solid feed intake than the calves fed the LF and MF treatments during the weaning period (523 g/d vs. 731 g/d and 696 g/d, $P < 0.01$; Table 2). Calves had lower solid feed intake in summer than winter during the preweaning (step-up: 28 g/d vs. 53 g/d, $P = 0.01$; peak: 64 g/d vs. 115 g/d, $P < 0.01$; weaning: 516 g/d vs. 785 g/d, $P < 0.01$) and the postweaning periods (2,827 g/d vs. 3,083 g/d, $P < 0.01$; Supplemental Table S1, see Notes).

The calves fed the HF treatment had lower NFC intake from solid feed than the calves fed the LF and MF treatments during the weaning period (230 g/d vs. 331 g/d and 321 g/d, $P < 0.01$). Calves had lower NFC intake from solid feed in summer than winter during the preweaning periods (step-up: 14 g/d vs. 24 g/d, $P = 0.02$; peak: 29 g/d vs. 49 g/d, $P = 0.02$; weaning: 239 g/d vs. 350 g/d, $P < 0.01$).

The calves fed the HF treatment had lower total CP intake than the calves fed the LF and MF treatments during the weaning period (301 g/d vs. 348 g/d and 341 g/d, $P < 0.01$). Calves had lower total CP intake in summer than

Table 2. Effects of fat content of high-protein milk replacer on DMI (g/d) during the step-up, peak, and weaning periods; LSM ± SEM

Item	Period	Treatment[1] Summer LF (n = 12)	Summer MF (n = 12)	Summer HF (n = 14)	Winter LF (n = 14)	Winter MF (n = 14)	Winter HF (n = 14)	SEM	P-value Treatment	P-value Season	P-value Treatment × season
Solid feed intake,[2] g/d											
	Step-up	23	38	24	44	62	53	10.8	0.30	0.01	0.93
	Peak	81	73	39	111	135	99	20.5	0.19	<0.01	0.68
	Weaning	608	567	373	855	826	673	60.8	<0.01	<0.01	0.90
NFC intake from solid feed,[3] g/d											
	Step-up	11	19	12	18	30	25	5.4	0.22	0.02	0.87
	Peak	36	34	17	44	61	43	10.1	0.22	0.02	0.56
	Weaning	281	270	167	381	374	294	30.7	<0.01	<0.01	0.90
Total CP intake,[4] g/d											
	Step-up	200	198	197	202	205	204	2.9	0.97	0.03	0.67
	Peak	349	342	332	353	360	353	5.1	0.17	<0.01	0.23
	Weaning	323	315	269	372	368	334	13.7	<0.01	<0.01	0.83
Total fat intake,[5] g/d											
	Step-up	139	177	213	136	175	212	1.2	<0.01	0.06	0.69
	Peak	239[c]	304[b]	361[a]	235[c]	304[b]	366[a]	1.9	<0.01	0.81	0.03
	Weaning	159	197	225	164	203	235	2.1	<0.01	<0.01	0.44

[a–c]Means with different superscripts are significantly different within a period ($P < 0.05$).

[1]LF was fed milk replacer containing 21% fat on DM basis, ME = 4.7 Mcal/kg; MF was fed milk replacer containing 26% fat on DM basis, ME = 5.0 Mcal/kg; HF was fed milk replacer containing 32% fat on DM basis, ME = 5.3 Mcal/kg.

[2]Solid feed intake was the sum of intake from calf starter and hay.

[3]The NFC intake was the sum of NFC intake from calf starter and hay.

[4]Total CP intake was the sum of CP intake from milk replacer, calf starter, and hay.

[5]Total fat intake was the sum of fat intake from milk replacer, calf starter, and hay.

winter during the preweaning (step-up: 198 g/d vs. 204 g/d, $P = 0.03$; peak: 341 g/d vs. 355 g/d, $P < 0.01$; weaning: 302 g/d vs. 358 g/d, $P < 0.01$) and the postweaning periods (630 g/d vs. 671 g/d, $P = 0.02$).

During the preweaning periods, the calves fed the HF treatment had higher total fat intake than the calves fed the MF treatment, and the calves fed the MF treatment had higher total fat intake than the calves fed the LF treatment (step-up: 213 g/d vs. 176 g/d vs. 138 g/d; peak: 364 g/d vs. 304 g/d vs. 237 g/d; weaning: 230 g/d vs. 200 g/d vs. 162 g/d, respectively, $P < 0.01$). Calves had lower total fat intake in summer than winter during the weaning (194 g/d vs. 200 g/d, $P < 0.01$) and the postweaning periods (98 g/d vs. 106 g/d, $P < 0.01$).

Plasma Metabolite Concentration

The calves fed the HF treatment had lower alkaline phosphatase concentration than the calves fed the MF treatment during the step-up period (281 IU/L vs. 355 IU/L, $P = 0.03$; Table 3). Calves had lower alkaline phosphatase concentration in summer than winter throughout the study (step-up: 283 IU/L vs. 357 IU/L; peak: 339 IU/L vs. 431 IU/L; weaning: 233 IU/L vs. 330 IU/L, respectively, $P < 0.01$). Calves had lower plasma alkaline phosphatase concentration in summer than winter during the postweaning period (276 IU/L vs. 352 IU/L, $P < 0.01$, Supplemental Table S2, see Notes).

The calves fed the HF and MF treatments had higher plasma triglyceride concentration than the calves fed the LF treatment during the step-up and peak periods (step-up: 6.7 mg/dL and 15.5 mg/dL vs. 10.5 mg/dL; peak: 40.4 mg/dL and 34.8 mg/dL vs. 23.6 mg/dL, respectively, $P < 0.01$) and the calves fed the HF treatment had higher concentration than the calves fed the LF treatment during the weaning period (30.1 mg/dL vs. 20.5 mg/dL, $P = 0.01$). Calves had higher plasma triglyceride concentration in summer than winter during the postweaning period (20.9 mg/dL vs. 17.5 mg/dL, $P < 0.01$).

The calves fed the HF treatment had higher plasma total cholesterol concentration than the calves fed the LF treatment during the step-up and peak periods (step-up: 89 mg/dL vs. 77 mg/dL, $P = 0.03$; peak: 163 mg/dL vs. 128 mg/dL, $P < 0.01$) and the calves fed the HF treatment had higher concentration than the calves fed the MF and LF treatments during the weaning period (170 mg/dL vs. 149 mg/dL and 131 mg/dL, $P < 0.01$). Calves had higher plasma total cholesterol concentration in summer than winter during the preweaning period (step-up: 89 mg/dL vs. 81 mg/dL, $P = 0.03$; peak: 159 mg/dL vs. 133 mg/dL, $P < 0.01$; weaning: 159 mg/dL vs. 141 mg/dL, $P = 0.01$). Calves had lower plasma total cholesterol concentration in summer than winter during the postweaning period (62 mg/dL vs. 78 mg/dL, $P < 0.01$).

The calves fed the HF treatment had higher plasma fatty acids concentration than the calves fed the LF treatment during the step-up period (307 mEq/L vs. 234 mEq/L, $P = 0.02$), and the calves fed the HF treatment had higher concentration than the calves fed the MF treatment and the calves fed the MF treatment had higher concentration than the calves fed the LF treatment during the peak and weaning periods (peak; 390 mEq/L vs. 282 mEq/L vs. 177 mEq/L; weaning: 423 mEq/L vs. 316 mEq/L vs. 240 mEq/L, respectively, $P < 0.01$). Calves had lower plasma fatty acids concentration in summer than winter during the postweaning period (109 mEq/L vs. 125 mEq/L, $P = 0.01$). We found an interaction of treatment and season for plasma fatty acids concentrations during the weaning period ($P < 0.05$); the calves fed the HF treatment had higher plasma fatty acids concentration than the calves fed the MF treatment and the calves fed the MF treatment had higher concentration than the calves fed the LF treatment in summer (419 mEq/L vs. 330 mEq/L vs. 203 mEq/L, $P < 0.01$); however, the calves fed the HF treatment had higher plasma fatty acids concentration than the calves fed the MF and LF treatments in winter (408 mEq/L vs. 303 mEq/L and 277 mEq/L, $P < 0.01$).

The calves fed the HF treatment had higher plasma albumin concentration than the calves fed the MF treatment during the weaning period (3.36 g/dL vs. 3.26 g/dL, $P = 0.04$). Calves had lower plasma albumin concentration in summer than winter during the preweaning periods (step-up: 2.77 g/dL vs. 3.04 g/dL; peak: 2.96 g/dL vs. 3.32 g/dL; weaning: 3.11 g/dL vs. 3.49 g/dL, respectively, $P < 0.01$). Calves had lower plasma albumin concentration in summer than winter during the postweaning period (3.17 g/dL vs. 3.62 g/dL, $P < 0.01$).

There was no effect of treatment on plasma protein concentration throughout the study. Calves had lower plasma total protein concentration in summer than winter during the preweaning periods (step-up: 5.28 g/dL vs. 6.06 g/dL; peak: 5.17 g/dL vs. 5.84 g/dL; weaning: 5.56 g/dL vs. 6.15 g/dL, respectively, $P < 0.01$). Calves had lower plasma total protein concentration in summer than winter during the postweaning period (6.00 g/dL vs. 6.58 g/dL, $P < 0.01$).

The calves fed the HF treatment had lower plasma urea nitrogen concentration than the calves fed the LF treatment during the peak and postweaning periods (peak: 8.8 mg/dL vs. 10.1mg/dL, $P = 0.01$; postweaning: 11.7 mg/dL vs. 12.7 mg/dL, $P = 0.04$), and the calves fed the HF treatment had lower concentration than the calves fed the LF and MF treatments during the weaning period (9.6 mg/dL vs. 11.4 mg/dL and 10.9 mg/dL, $P < 0.01$). We found an interaction of treatment and season for plasma urea nitrogen concentrations during the step-up period ($P = 0.03$); treatment had no effect on plasma urea nitrogen concentration in summer, and the concentration was

Table 3. Effects of fat content of high-protein milk replacer on plasma metabolic and hormone concentrations (LSM ± SEM) during the step-up, peak, and weaning periods

		Treatment[1]							*P*-value		
		Summer			Winter						
		LF	MF	HF	LF	MF	HF				
Item	Period	(n = 12)	(n = 12)	(n = 14)	(n = 14)	(n = 14)	(n = 14)	SEM	Treatment	Season	Treatment × season
Alkaline phosphatase, IU/L											
	Step-up	254.6	326.2	267.5	393.0	384.1	293.6	28.19	0.03	<0.01	0.13
	Peak	326.7	363.8	325.5	499.6	440.2	354.4	33.97	0.07	<0.01	0.10
	Weaning	221.5	247.6	228.9	342.8	366.6	279.7	24.30	0.10	<0.01	0.25
Triglyceride, mg/dL											
	Step-up	11.3	15.4	17.1	9.7	15.6	16.3	1.62	<0.01	0.59	0.85
	Peak	28.2	34.3	42.5	19.1	35.2	38.3	4.24	<0.01	0.24	0.51
	Weaning	19.2	23.1	31.7	21.9	26.8	28.5	3.08	0.01	0.68	0.48
Total cholesterol, mg/dL											
	Step-up	75.8	84.5	81.2	78.5	93.6	94.9	4.76	0.03	0.03	0.51
	Peak	120.8	139.1	138.0	134.7	155.0	187.5	8.70	<0.01	<0.01	0.07
	Weaning	127.6	139.4	156.7	134.4	158.0	183.4	7.69	<0.01	0.01	0.43
Fatty acids, mEq/L											
	Step-up	242.5	292.9	316.6	224.7	263.8	297.0	24.96	0.02	0.28	0.97
	Peak	169.8	293.6	409.2	183.2	270.1	371.0	22.28	<0.01	0.38	0.49
	Weaning	202.6	330.1	418.9	276.6	302.9	407.6	21.50	<0.01	0.50	0.05
Albumin, g/dL											
	Step-up	2.8	2.8	2.8	3.0	3.1	3.0	0.04	0.37	<0.01	0.19
	Peak	2.9	3.0	3.0	3.3	3.3	3.3	0.03	0.46	<0.01	0.98
	Weaning	3.1	3.1	3.1	3.5	3.4	3.6	0.04	0.04	<0.01	0.41
Total protein, g/dL											
	Step-up	5.3	5.3	5.3	6.0	6.2	6.0	0.19	0.80	<0.01	0.67
	Peak	5.2	5.2	5.2	5.7	6.0	5.8	0.10	0.52	<0.01	0.68
	Weaning	5.6	5.6	5.6	6.1	6.2	6.2	0.10	0.98	<0.01	0.89
Urea nitrogen, mg/dL											
	Step-up	11.2	10.5	12.1	12.0	11.9	10.3	0.64	0.78	0.78	0.03
	Peak	9.6	9.3	8.9	10.5	10.0	8.7	0.40	0.01	0.16	0.28
	Weaning	10.3	10.1	9.2	12.4	11.6	10.1	0.43	<0.01	<0.01	0.37
Glucose, mg/dL											
	Step-up	100.6	101.8	97.0	109.4	109.9	105.7	2.76	0.23	<0.01	0.99
	Peak	116.5	115.5	107.7	130.3	129.4	129.5	3.40	0.32	<0.01	0.39
	Weaning	101.1	102.3	104.2	123.8	117.8	122.9	2.25	0.29	<0.01	0.29
Total ketone body, µmol/L											
	Step-up	58.1	57.5	67.6	59.3	84.9	72.5	8.34	0.26	0.10	0.25
	Peak	56.8	69.5	79.6	69.0	107.7	99.7	11.90	0.05	0.02	0.54
	Weaning	130.0	143.2	127.0	171.4	159.9	193.6	11.77	0.66	<0.01	0.11
IGF-1, ng/mL											
	Step-up	43.9	44.9	45.6	52.2	46.4	47.0	3.66	0.80	0.22	0.56
	Peak	82.9	82.5	88.6	100.7	98.8	98.8	6.06	0.88	<0.01	0.80
	Weaning	77.2	75.1	71.5	92.9	89.6	87.5	6.13	0.67	<0.01	0.99
Insulin, ng/mL											
	Step-up	6.2	4.6	6.6	4.3	4.3	4.4	0.50	0.10	<0.01	0.13
	Peak	10.6	7.5	8.9	4.6	5.8	4.8	0.64	0.34	<0.01	0.01
	Weaning	8.0	5.2	7.9	4.2	4.7	4.2	0.63	0.14	<0.01	0.02

[1]LF was fed milk replacer containing 21% fat on DM basis, ME 4.7 Mcal/kg; MF was fed milk replacer containing 26% fat on DM basis, ME 5.0 Mcal/kg; HF was fed milk replacer containing 32% fat on DM basis, ME 5.3 Mcal/kg.

affected by treatment in winter ($P = 0.07$, Figure 1B); however, there were no significant differences between treatment. We found an interaction of treatment and season for plasma urea nitrogen concentrations during the postweaning period ($P < 0.01$); treatment had no effect on plasma urea nitrogen concentration in summer; however, the calves fed the HF treatment had lower plasma urea nitrogen concentration than the calves fed the LF treatment in winter (12.3 mg/dL vs. 14.6 mg/dL, $P < 0.01$).

There was no effect of treatment on plasma glucose concentration throughout the study. Calves had lower plasma glucose concentration in summer than winter during the preweaning periods (step-up: 100 mg/dL vs. 108 mg/dL; peak: 113 mg/dL vs. 130 mg/dL; weaning: 103 mg/dL vs. 122 mg/dL, respectively, $P < 0.01$). Calves had lower plasma glucose concentration in summer than winter during the postweaning period ($P < 0.01$, 92 vs. 110 mg/dL).

The calves fed the MF and HF treatments tended to have higher plasma total ketone body concentration (89 µmol/L and 90 µmol/L) than the calves fed the LF treatment (63 µmol/L) during the peak period. Calves had lower plasma total ketone body concentration in summer than winter during the peak and weaning periods (peak: 69 µmol/L vs. 92 µmol/L, $P = 0.02$; weaning: 133 µmol/L vs. 175 µmol/L, $P < 0.01$). Calves had lower plasma total ketone body concentration in summer than winter during the postweaning period (363 µmol/L vs. 426 µmol/L, $P < 0.01$).

Plasma Hormone Concentration

There was no effect of treatment on plasma IGF-1 concentration throughout the study. Calves had lower plasma IGF-1 concentration in summer than winter during the peak and weaning periods (peak: 85 ng/mL vs. 99 ng/mL; weaning: 75 ng/mL vs. 90 ng/mL, respectively, $P < 0.01$). Calves had lower plasma IGF-1 concentration in summer than winter during the postweaning period (103 ng/mL vs. 117 ng/mL, $P = 0.03$).

Treatment had no effect on plasma insulin concentration throughout the study. Calves had higher plasma insulin concentration in summer than winter during the preweaning periods (step-up: 5.81 ng/mL vs. 4.35 ng/mL; peak: 9.01 ng/mL vs. 5.05 ng/mL; weaning: 7.02 ng/mL vs. 4.37 ng/mL, respectively, $P < 0.01$). Calves had higher plasma insulin concentration in summer than winter during the postweaning period (6.16 ng/mL vs. 4.69 ng/mL, $P < 0.01$). We found an interaction of treatment and season for plasma insulin concentration during the peak period ($P = 0.01$); plasma insulin concentration was not affected by treatment in summer, but the calves fed the MF treatment had higher plasma insulin concentration than the calves fed the LF treatment in winter (5.8 ng/mL vs. 4.6 ng/mL, $P = 0.02$). We also found an interaction of treatment and season for plasma insulin concentration during the weaning period ($P = 0.02$); the calves fed the MF treatment tended to have lower plasma insulin concentration than the calves fed the LF and HF treatments (5.2 ng/mL vs. 8.0 ng/mL and 7.9 ng/mL), but was not affected by treatment in winter.

DISCUSSION

Effects of Fat Content of Milk Replacer on Lipid Metabolites

In our study, HF calves had higher plasma cholesterol concentrations during the preweaning periods. Our results are consistent with a study conducted by Echeverry-Munera et al. (2021), in which calves fed a high-fat milk replacer had higher serum cholesterol concentration than calves fed a low-fat milk replacer during preweaning period. High fat intake is associated with high blood cholesterol in human medicine (Siri-Tarino et al., 2010). In contrast to the negative perceptions associated with high blood cholesterol in humans, high blood cholesterol concentrations have positive effects on health and reduced early mortality in calves (Renaud et al., 2018). Additionally, fatty acid intake, especially linolenic acid, enhanced immune function of young calves (Hill et al., 2011), and feeding high-fat milk replacer to calves may have positive health effects. In fact, fat intake from liquid feeds is associated with decreased mortality (Urie et al., 2018), improved fecal scores (Amado et al., 2019), and reduced therapeutic interventions (Berends et al., 2020) in dairy calves. However, we did not observe reductions in diarrhea (Fukami et al., 2025) or other health benefits for calves consuming greater linolenic acid from milk replacers with greater fat content.

Wilms et al. (2024c) reported that calves fed milk replacer (24% CP and 30% fat) containing mainly rapeseed oil had higher plasma cholesterol concentration, particularly in the form of low-density lipoprotein (**LDL**) cholesterol. They suggested that this result can be attributed to high PUFA, accounting for 12.8% of total fatty acids in the milk replacer (Wilms et al., 2024c). We used vegetable oil containing PUFA at 8.5% of total fatty acid. Although the concentration of PUFA was not as high as that in the study by Wilms et al. (2024c), our study suggested that increasing the fat content of milk replacer from vegetable oil would increase plasma total cholesterol concentrations. However, we did not measure LDL cholesterol in the current study, and its effects on health of dairy calves warrant further investigation.

In our study, greater fat content of milk replacer increased plasma fatty acid concentrations, which is consistent with previous studies (Tikofsky et al., 2001; Berends et al., 2020; Echeverry-Munera et al., 2021). After milk consumption, dietary triglyceride is hydrolyzed by lipase into

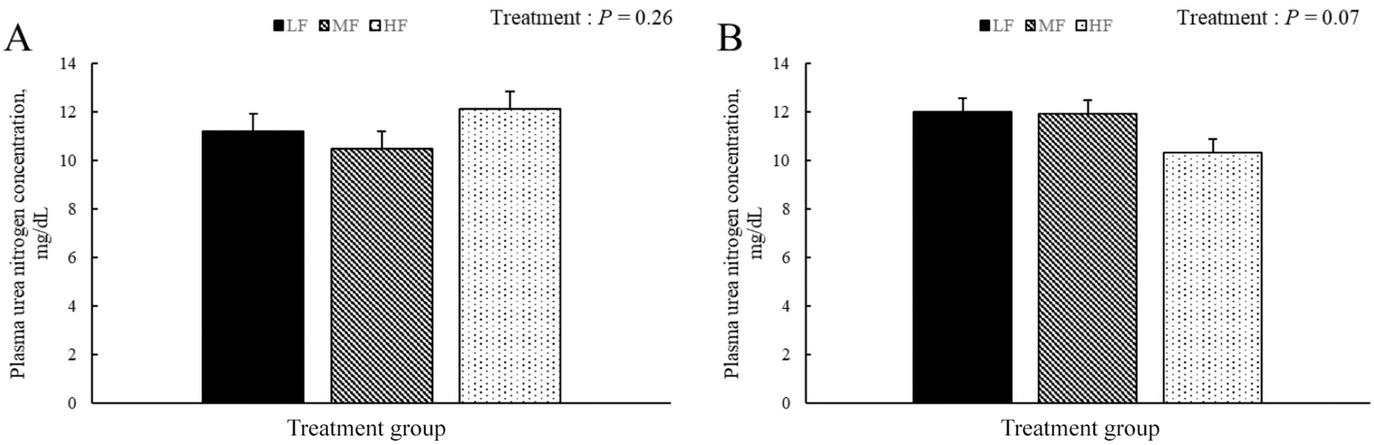

Figure 1. Effects of fat content of milk replacer on urea nitrogen concentrations during the step-up period (A) in summer, and (B) in winter. The LF group was fed milk replacer containing 21% fat on DM basis, the MF group was fed milk replacer containing 26% fat on DM basis, and the HF group was fed milk replacer containing 32% fat on DM basis. Error bars represent SEM.

fatty acids and monoacylglycerol, which are subsequently absorbed by enterocytes (Lambert and Parks, 2012; Wilms et al., 2024a). Within the enterocyte, fatty acids undergo various processes including incorporation into cholesteryl ester or phospholipid, oxidation, and re-esterification to form triglyceride for incorporation into chylomicrons or storage in an intracellular triglyceride storage pool (Lambert and Parks, 2012; Wilms et al., 2024a). However, short-chain and medium-chain fatty acids can be solubilized in the aqueous phase of the intestinal contents, bound to albumin for absorption, and transported to the liver via the portal vein (Decker, 1996; Ramírez et al., 2001). Increased plasma fatty acid concentration for calves fed milk replacers of greater fat content may indicate that fatty acid supply exceeded the metabolic capacity of the liver and released without incorporation into triglycerides.

In addition, the calves fed the MF and HF treatments tended to have higher plasma total ketone body concentration than the calves fed the LF treatment, and it is likely attributed to higher plasma fatty acids concentrations for MF and HF compared with LF calves. Wilms et al. (2022) also reported that calves fed high-fat milk replacer had higher blood fatty acids and BHB concentrations than calves fed low-fat milk replacer. Moreover, plasma total ketone body concentration including BHB synthesized from fatty acids in the liver (Newman and Verdin, 2017), also increased as the fat content of milk replacer increased. A previous study showed that increasing serum BHB concentrations is a good indicator of rumen development, and thus, of a shift in nutrient source from liquid to solid feed (Coverdale et al., 2004). However, it is unlikely that higher plasma total ketone body concentration in MF and HF calves during the peak period came from greater rumen development, as intake of solid feeds was similar across treatment groups.

Effects of Fat Content of Milk Replacer and Season on Nitrogen Metabolites

The calves fed the HF treatment had lower plasma urea nitrogen concentration than the calves fed the LF treatment after the peak period. Previous studies reported that increasing the fat content of milk replacer decrease BUN concentration (Berends et al., 2020), consistent with our study. Urea is the end product of protein catabolism and the major solute in urine (Bankir et al., 1996). Greater protein intake is a contributing factor for greater concentration of plasma urea nitrogen (Wilms et al., 2024b), but the 3 milk replacers in our study had equal CP content, and total CP intake was not affected by treatment during the peak period. However, plasma urea nitrogen concentration decreased linearly as the fat content of milk replacers increased. Calves with greater ME intake (Fukami et al., 2025; 5.6 Mcal/d in LF, 6.0 Mcal/d in MF, 6.2 Mcal/d in HF, $P < 0.01$) may increase nitrogen use efficiency in our study; when more energy was available for body protein synthesis, less protein might have been lost as urea nitrogen, which indicates greater nitrogen utilization efficiency (Kohn et al., 2005). In fact, there was a negative relationship between total ME intake and plasma urea nitrogen concentration during the peak period ($P < 0.01$; $r = -0.21$).

During the weaning period, the calves fed the HF had lower total CP intake than the calves fed the LF and MF treatments (301 g/d vs. 348 and 341 g/d, $P < 0.01$) due to lower solid feed intake. As such, lower plasma urea nitrogen concentration in calves fed the HF treatment during the weaning period can be attributed to lower total CP intake for the calves fed the HF treatment. Plasma urea nitrogen concentration during the weaning period was positively related to total CP intake ($P < 0.01$; $r = 0.42$). Furthermore, greater NFC intake is expected to decrease plasma urea nitrogen concentration as it provides energy for efficient microbial protein synthesis (Shabi et al., 1998; Wei et al., 2021). Maximizing microbial protein synthesis enhances nitrogen use efficiency and reduces nitrogen excretion in urine (Thomas, 1973; Wei et al., 2021). However, in our study, the calves fed the HF treatment had lower NFC intake from solid feed than the calves fed the LF and MF treatments during the weaning period (230 g/d vs. 331 and 321 g/d, $P < 0.01$). Therefore, plasma urea nitrogen concentration was mainly related to total CP intake. It should be studied in detail in the future including nitrogen retention calculated by nitrogen intake, fecal nitrogen, and urea nitrogen.

We observed an interaction of treatment and season for plasma urea nitrogen concentration during the step-up period; plasma urea nitrogen concentrations were not affected by treatment in summer but tended to decrease linearly in winter as the fat content of milk replacers increased although total CP intake was not different among the treatments. This may be due to the higher maintenance energy requirement in winter compared with summer in this study. The thermoneutral zone for young calves is 15°C to 25°C from birth to 3 wk of age, and the lower critical temperature drops to 10°C after 3 wk of age (NASEM, 2021). Maintenance energy requirement increases by 2.01 kcal/kg$^{0.75}$ per day for each degree decreasing in environmental temperature (°C) below the lower critical temperature (NASEM, 2021). In our study, based on the calculation using NASEM (2021), calves needed to use 230 kcal/d more energy for maintenance during the step-up period in winter than summer (Fukami et al., 2025). When maintenance energy requirements are high, less energy is available for body protein synthesis, and more protein is lost as urea nitrogen. We reported that withers height gain during the step-up period decreased linearly in winter as the fat content of milk replacer decreased, but not in summer (Fukami et al., 2025). We previously reported that energy available for growth was lower in winter than summer (1.38 Mcal/d vs. 1.57 Mcal/d, $P < 0.01$; Fukami et al., 2025) and it increased as the fat content of milk replacer increased (1.67 Mcal/d in HF, 1.49 Mcal/d in MF, 1.26 Mcal/d in LF, $P < 0.01$). Decreased plasma urea nitrogen concentration in winter, as the fat content of milk replacer increased, suggests greater nitrogen use efficiency and is consistent with increased withers height gain during the step-up period in winter (Fukami et al., 2025).

Effects of Fat Content of Milk Replacer and Season on Hormones

We found interactions of treatment and season for plasma insulin concentration during the peak and weaning periods. During the peak period, the calves fed the MF treatment had higher insulin concentration than the calves fed the LF treatment in winter, but insulin concentration was not affected by treatment in summer. In contrast, during the weaning period, the concentration tended to be lower in the calves fed the MF treatment than the calves fed the LF and HF treatments but was not affected by treatment in summer.

It is possible that both lactose and fat in milk replacers affect insulin concentration. Negative effects on glucose homeostasis (hyperglycemia, glucosuria, and hyperinsulinemia) were reported as a result of high lactose inclusion (42%–50% DM) in milk replacer (Hugi et al., 1998; Echeverry-Munera et al., 2021). High fat feeding was shown to increase pancreatic responsiveness in rodents with glucose-induced hypersecretion of insulin; however, this effect was lost after 2 mo (Cruciani-Guglielmacci et al., 2005). In addition, heat stress increases insulin secretion and sensitivity (O'Brien et al., 2010; Fontoura et al., 2023). The interaction between treatment and season during the peak period in our study can be explained as follows; in summer, insulin concentration increased regardless of the energy source of milk replacer, indicating that environmental factors played a major role. In winter, insulin concentration might have varied depending on the energy source, MF milk replacer with a 35:26 ratio of lactose to fat had a higher insulin concentration than LF milk replacer with a 40:21 ratio. However, the similar treatment effects were not observed during the weaning period possibly because of decreased milk replacer intake.

CONCLUSIONS

Increasing the fat content of milk replacer decreased plasma urea nitrogen concentration during the peak and weaning periods regardless of season, possibly due to lower total CP intake, but only during the step-up period in winter. In addition, increasing the fat content of milk replacer increased plasma cholesterol and fatty acid concentrations, but did not exert consistent effects on plasma insulin concentration. These results suggest that increasing the fat content of milk replacer may improve nitrogen efficiency, and this should be considered in milk replacer formulation particularly for winter when maintenance energy requirement increases.

NOTES

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Nonstandard abbreviations used: EBW = empty body weight, HF = milk replacer containing 32% fat; LDL = low-density lipoprotein; LF = milk replacer containing 21% fat; MF = milk replacer containing 26% fat.

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